

Birla Institute of Technology & Science, Pilani
Work Integrated Learning Programmes Division
Second Semester 2025-2026

Mid-Semester Test
(EC-2 Regular)

Course No. : AIMLZG511
Course Title : Deep Neural Network
Nature of Exam : Closed Book
Weightage : 30%
Duration : 2 Hours
Date of Exam : 14/03/2026 (FN/AN)

No. of Pages	= 02
No. of Questions	= 05

Note to Students:

1. Please follow all the *Instructions to Candidates* given on the cover page of the answer book.
2. All parts of a question should be answered consecutively. Each answer should start from a fresh page.
3. Assumptions made if any, should be stated clearly at the beginning of your answer.
4. Use of Scientific Calculator is permitted.

Q1 A binary classification model produces the following outputs for 10 observations: **(02+02+02=06Marks)**

Predicted Probabilities: 0.90, 0.80, 0.70, 0.60, 0.55, 0.40, 0.30, 0.20, 0.10, 0.05

True Class Labels: 1, 1, 0, 0, 1, 0, 0, 0, 0, 0

- (a) At threshold = 0.50, compute the confusion matrix, Precision, Recall, and F1-score.
- (b) At threshold = 0.30, compute the confusion matrix, Precision, Recall, and F1-score.
- (c) Which threshold is preferable and why for (i) credit risk and (ii) disease detection?
Briefly justify why 0.50 may not be optimal in imbalanced data.

Q2 Design a 2-layer perceptron network using multiple AND/OR gates **(01+03+01=05 Marks)**

to implement the Boolean function $F(x_1, x_2, x_3, x_4) = (x_1 \wedge x_2) \vee (x_3 \wedge x_4)$

where $x_1, x_2, x_3, x_4 \in \{0, 1\}$ and

Activation function for each neuron is $f(z) = 1$ if $z \geq 0$,

$f(z) = 0$ if $z < 0$

- a) Decompose the Boolean function into hidden layer and output layer operations.
- b) Derive the inequalities for an AND, OR gates using the perceptron model and determine suitable weights and bias.
- c) Show the final weights, biases, and draw the 2-layer perceptron network.

Q3. Consider a 2-layer neural network: **(02+02+02+02=08 Marks)**

2 inputs $\rightarrow$ 3 hidden units (ReLU) $\rightarrow$ 2 outputs (Sigmoid).

Given a single input x and parameters $W[1]$, $b[1]$, $W[2]$, $b[2]$, perform one complete forward and backward pass for a single training example.

Input: $x = [1.0, 0.5]^T$, Target: $y = [1, 0]^T$, Hidden activation: ReLU; Output activation: Sigmoid, Loss: Binary Cross-Entropy (BCE)

Parameters: $W[1] = [[0.5, 0.2], [-0.3, 0.8], [0.1, -0.4]]$ (shape 3×2), $b[1] = [0.1, -0.2, 0.3]^T$ (shape 3×1)
 $W[2] = [[0.4, -0.1, 0.6], [0.2, 0.7, -0.3]]$ (shape 2×3), $b[2] = [0.2, -0.1]^T$ (shape 2×1)

- a) Compute $z[1] = W[1]x + b[1]$ and $a[1] = \text{ReLU}(z[1])$.

- b) Compute $z[2] = W[2]a[1] + b[2]$ and $a[2] = \sigma(z[2])$.
- c) Using Binary Cross-Entropy loss with a given target y , compute the output-layer error signal $\delta[2]$ and gradients $dW[2]$, $db[2]$.
- d) Backpropagate to the hidden layer and explain what happens to the gradient for any hidden neuron where $z[1] \leq 0$ under ReLU.

Q4 Given data: (1,2), (2,5), (4,11), (5,14).

(02+02+02=06Marks)

Model: $\hat{y} = w_0 + w_1x$

Loss: $J = \frac{1}{2N} \sum (y - \hat{y})^2$

$\eta=0.05, w_0=1, w_1=-1$

- (a) Compute gradient at initialization.
- (b) Perform one batch update. (Show intermediate residual computation clearly.)
- (c) Find optimal weights using normal equations and state whether the first update moves toward optimum.

Q5 You are designing a deep feedforward neural network (DFNN) to predict house prices. **(01*5=05 Marks)**

Dataset: 500 samples, 20 numerical features, regression task.

Two architectures are tried:

ModelA: 6 hidden layers, 512 units each. Training loss decreases to near zero, validation loss increases significantly.

ModelB: 3 hidden layers, 64, 32, 16 units (funnel architecture). Training loss moderate, validation loss low.

Tasks:

- (a) Diagnose Model A and name the phenomenon occurring.
- (b) Suggest at least three practical architectural changes to improve validation performance.
- (c) Compute the number of parameters in the first hidden layer of Model A (Include bias).
- (d) Compare the total number of parameters (order of magnitude) between Model A and Model B.
- (e) Briefly explain why funnel architectures (like Model B) are often preferred when data is limited.